LAB 1 – REQUIREMENTS ENGINEERING & UML USE-CASE MODELLING

Vaccination Cohort & Dose Scheduling System

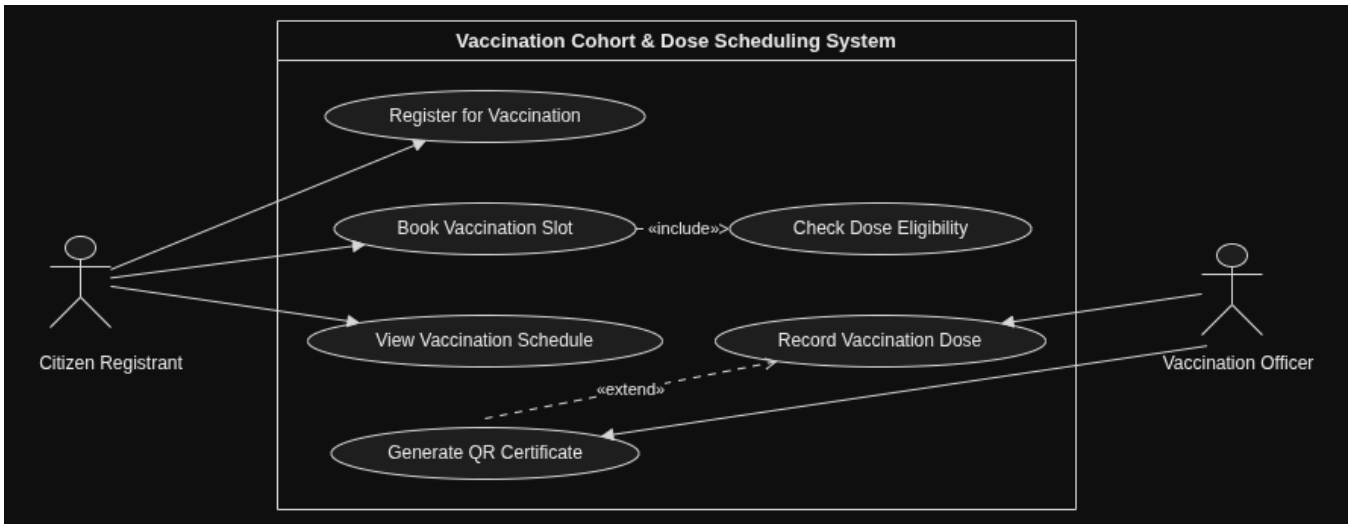
PES2UG24CS454 | SHALIK S

1. Requirements Table

The system is used to organize vaccination registration, check dose intervals, book vaccination center slots, record doses, and generate QR vaccination certificates.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall enforce dose interval rules before allowing a citizen to book the next dose.	High	Pass: Dose 2 is blocked when the required interval (for example, 28 days) is not completed. Fail: An early slot is confirmed.	Makes sure the required gap between doses is followed.	Need to clearly define the minimum interval.
FR-002	Functional	The system shall allow a citizen registrant to register for a vaccination cohort using the required details.	High	Pass: Valid details create a registration record. Fail: Incomplete required details are accepted.	Needed to keep citizen vaccination records.	Specify which details are required.
FR-003	Functional	The system shall allow an eligible citizen to view and book an available vaccination center slot.	High	Pass: Available slots are shown and one selected slot is confirmed. Fail: A full or already booked slot is confirmed.	Helps citizens get a vaccination time and place.	Check that one slot cannot be booked twice.
FR-004	Functional	The system shall allow a vaccination officer to record a dose given to a registered citizen.	High	Pass: The dose number, date, citizen, and center are saved. Fail: An invalid dose record is saved.	Keeps the vaccination history updated.	Include dose number and vaccination date.
FR-005	Functional	The system shall generate a QR vaccination certificate after a valid vaccination record is completed.	High	Pass: A QR certificate is generated from the citizen's valid vaccination record. Fail: A certificate is generated without a valid record.	Gives the citizen a digital vaccination certificate.	Certificate should only be generated after vaccination.
NFR-001	Nonfunctional	The vaccination certificate verification endpoint shall authenticate digitally signed QR codes offline/online in under 150 ms.	High	Pass: Tests show verification under 150 ms and the QR signature is accepted only when valid. Fail: The time or security check fails.	Makes certificate checking fast and secure.	Check the 150 ms limit during testing.
NFR-002	Nonfunctional	The system shall be available for at least 99% of the defined normal operating period.	Medium	Pass: Availability testing records at least 99% uptime for the test period. Fail: Uptime is below 99%.	The system should be dependable when people need to book or check records.	Define the period used for the uptime test.

2. UML Use-Case Diagram



.

3. Use-Case Flow Specification

Use Case: Book Vaccination Slot

Primary Actor	Citizen Registrant
Preconditions	The citizen is registered, is eligible for the required dose, and vaccination center slots are available.
Postconditions	The selected slot is booked for the citizen and the vaccination schedule is updated.

Main Success Scenario

1. The citizen selects Book Vaccination Slot.
2. The system checks the citizen's vaccination record.
3. The system checks the required dose interval and eligibility.
4. The system shows the available vaccination center slots.
5. The citizen selects a suitable date, time, and center.
6. The system checks that the selected slot is still available.
7. The system books the slot and updates the citizen's schedule.
8. The system shows the booking confirmation.

Alternate Flow – Dose Interval Not Completed

9. At Step 3, the system finds that the required interval since the previous dose has not been completed.
10. The system does not allow the citizen to book the next dose.
11. The system shows the reason and the earliest date when the citizen can book.
12. No vaccination slot is reserved.